Spectral decomposition

Suppose $A \in M_{n\times n}$ is symmetric and $A^T = A$.

$$A = PDP^{-1} = PDP^T \qquad P^T = P^{-1}$$

where $D = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}$ is diagonal and

$P = [u_1 \cdots u_n]$ has columns that form an orthonormal basis for $\mathbb{R}^n$. Then

$$A = [u_1 \cdots u_n] \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix} \begin{bmatrix} u_1^T \\ \vdots \\ u_n^T \end{bmatrix} = \cdots = \lambda_1 u_1 u_1^T + \cdots + \lambda_n u_n u_n^T$$

- $\lambda_j u_j u_j^T$ is a symmetric matrix $\forall\ 1 \le j \le n$.
- $$\begin{aligned} Av &= \lambda_1 u_1 u_1^T v + \cdots + \lambda_n u_n u_n^T v \\ &= \lambda_1 \langle v, u_1 \rangle u_1 + \cdots + \lambda_n u_n \langle v, u_n \rangle \\ &= \lambda_1 \operatorname{proj}_{u_1} v + \cdots + \lambda_n \operatorname{proj}_{u_n} v. \end{aligned}$$

Example

Take $A = \begin{bmatrix} 1 & -2 \\ -2 & 1 \end{bmatrix}$.

$$\det(A - xI) = (1-x)^2 - 4 = x^2 - 2 - 3 = (x-3)(x+1)$$

$\det(A - xI) = 0 \iff x = 3$ or $x = -1$

Eigenvalues of A are 3 and -1.

$$\mathrm{Nul}(A-3I) = \cdots = \mathrm{span}\left\{\begin{bmatrix}-1\\1\end{bmatrix}\right\}$$

$$\mathrm{Nul}(A+I) = \cdots = \mathrm{span}\left\{\begin{bmatrix}1\\1\end{bmatrix}\right\}$$

Normalizing $\begin{bmatrix}-1\\1\end{bmatrix}$ and $\begin{bmatrix}1\\1\end{bmatrix}$ we obtain an orthonormal basis $\left\{\begin{bmatrix}-1/\sqrt{2}\\1/\sqrt{2}\end{bmatrix}, \begin{bmatrix}1/\sqrt{2}\\1/\sqrt{2}\end{bmatrix}\right\}$.

Set $P = \begin{bmatrix}-1/\sqrt{2} & 1/\sqrt{2}\\1/\sqrt{2} & 1/\sqrt{2}\end{bmatrix}$ and $D = \begin{bmatrix}3 & 0\\0 & -1\end{bmatrix}$.

Then $A = PDP^T$

$$= 3\begin{bmatrix}-1/\sqrt{2}\\1/\sqrt{2}\end{bmatrix}\begin{bmatrix}-1/\sqrt{2} & 1/\sqrt{2}\end{bmatrix} + (-1)\begin{bmatrix}1/\sqrt{2}\\1/\sqrt{2}\end{bmatrix}\begin{bmatrix}1/\sqrt{2} & 1/\sqrt{2}\end{bmatrix}$$

$$= 3\begin{bmatrix}(-1/\sqrt{2})^2 & (-1/\sqrt{2})(1/\sqrt{2})\\(1/\sqrt{2})(-1/\sqrt{2}) & (1/\sqrt{2})^2\end{bmatrix} + (-1)\begin{bmatrix}(1/\sqrt{2})^2 & (1/\sqrt{2})^2\\(1/\sqrt{2})^2 & (1/\sqrt{2})^2\end{bmatrix}$$

$$= 3\begin{bmatrix}1/2 & -1/2\\-1/2 & 1/2\end{bmatrix} - \begin{bmatrix}1/2 & 1/2\\1/2 & 1/2\end{bmatrix}.$$

7.2 Quadratic Forms

In section $A \in M_{n\times n}$ and A is symmetric i.e. $A = A^T$.

Def. The <u>quadratic form</u> corresponding to $A = A^T$ is the function $Q: \mathbb{R}^n \to \mathbb{R}$, defined

$$Q(x) = x^T A x, \qquad \forall\, x \in \mathbb{R}^n$$

(maybe it Q_A would be better notation).

Remarks

- Recall that the standard inner product on $\mathbb{R}^n$ satisfies $\langle u, v \rangle = u^T v \quad \forall\, u, v \in \mathbb{R}^n$.

$$\Rightarrow \quad \langle x, x \rangle = x^T x \qquad \forall\, x \in \mathbb{R}^n$$
$$= x^T I x$$

We can recover the quadratic form corresponding to the identity matrix by taking the same vector in both "slots".

Example

Take $n = 2$.

The quadratic form for $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ is defined as $[x_1 \ x_2] \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = x_1^2 + x_2^2$

$\forall \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \in \mathbb{R}^2$.

Take $A = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$. Then $A^T = A$.

and its quadratic form is given by

$$[x_1 \ x_2] \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = [x_1 \ x_2] \begin{pmatrix} \lambda_1 x_1 \\ \lambda_2 x_2 \end{pmatrix}.$$

$$= \lambda_1 x_1^2 + \lambda_2 x_2^2.$$

Take $A = \begin{bmatrix} a & b \\ b & c \end{bmatrix} \in M_{2\times 2}$. Then $A^T = A$

and its quadratic form is given by

$$(x_1 \ x_2) \begin{bmatrix} a & b \\ b & c \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = (x_1 \ x_2) \begin{bmatrix} ax_1 + bx_2 \\ bx_1 + cx_2 \end{bmatrix}$$

$$= x_1(ax_1 + bx_2) + x_2(bx_1 + cx_2)$$

$$= a x_1^2 + b x_1 x_2 + b x_1 x_2 + c x_2^2$$

$$= a x_1^2 + 2b x_1 x_2 + c x_2^2.$$

This is a polynomial in x_1, x_2 which is homogeneous of degree 2. (All monomials have degree 2).

Let's call the quadratic form attached to $\begin{bmatrix} 3 & -2 \\ -2 & 7 \end{bmatrix}$ Q.

By the above $Q\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = 3x_1^2 + 2(-2)x_1x_2 + 7x_2^2$

$$= 3x_1^2 - 4x_1x_2 + 7x_2^2$$

$$Q\left(\begin{bmatrix} -1 \\ 1 \end{bmatrix}\right) = 3(-1)^2 - 4(-1)(1) + 7(1)^2$$

$$= 3 + 4 + 7 = 14$$

Example 2 in textbook.

Take $A = \begin{bmatrix} 5 & -1/2 & 0 \\ -1/2 & 3 & 4 \\ 0 & 4 & 2 \end{bmatrix}$. Then $A^T = A$.

Then $Q\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} 5 & -1/2 & 0 \\ -1/2 & 3 & 4 \\ 0 & 4 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

$$= \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} 5x_1 - 1/2 x_2 \\ -1/2 x_1 + 3x_2 + 4x_3 \\ 4x_2 + 2x_3 \end{bmatrix}$$

$$= \cdots$$

$$= 5x_1^2 + 3x_2^2 + 2x_3^2 - x_1x_2 + 8x_2x_3$$

- In general a quadratic form is a homogeneous polynomial in the entries of a vector of degree 2. The diagonal entries of A match the entries of the "square" monomials and the coefficients of the "mixed" monomials match the entries of the matrix with a factor of two.

- Using these observations we can convert any homogeneous polynomial into a quadratic form.

  e.g. The homogeneous polynomial in x_1, x_2, x_3

$3x_1^2 + 2x_2^2 + x_3^2 + x_1x_2 + (-2)x_2x_3 + 5x_1x_3$

corresponds to the quadratic form of the symmetric matrix $A = \begin{bmatrix} 3 & 1/2 & 5/2 \\ 1/2 & 2 & -1 \\ 5/2 & -1 & 1 \end{bmatrix}$.

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symmetric matrices $\longleftrightarrow$ quadratic forms $\longleftrightarrow$ homogeneous polynomials of degree 2.

Example.

If $A = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}$ is diagonal then

$$Q\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \lambda_1 x_1^2 + \dots + \lambda_n x_n^2 .$$

Recollections from chapter 4.

Let $\mathcal{E} = \{e_1, \dots, e_n\}$ be the standard basis for $\mathbb{R}^n$

i.e. $e_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$, $e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$, etc.

Take $n = 3$. $v = \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix} \in \mathbb{R}^3$.

Then $v = \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix} = (-1)\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + 2\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + 3\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

$= (-1)e_1 + 2e_2 + 3e_3$

$\Rightarrow [v]_{\mathcal{E}} = \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix} = v.$

Similarly $[v]_{\mathcal{E}} = v$ for any $v \in \mathbb{R}^3$.

In general $[v]_{\mathcal{E}} = v$. for any $v \in \mathbb{R}^n$.

Now let $\mathcal{B} = \{u_1, \ldots, u_n\}$ be any basis for $\mathbb{R}^n$.

Then $v = [v]_{\mathcal{E}} = \underset{\mathcal{E} \leftarrow \mathcal{B}}{P} [v]_{\mathcal{B}}$ (*)

where $\underset{\mathcal{E} \leftarrow \mathcal{B}}{P} = \left[[u_1]_{\mathcal{E}} \cdots [u_n]_{\mathcal{E}} \right]$ (Thm 15 §4.6)

$= \left[u_1 \cdots u_n \right]$

Recall from Thm 2 that the symmetric matrix $A \in M_{n \times n}$ is orthogonally diagonalizable, i.e.

$A = PDP^T$ where.

$D = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}$ is diagonal, $P^T = P^{-1}$,

and $P = [u_1 \cdots u_n]$ where $\{u_1, \ldots, u_n\}$ is an orthonormal basis for $\mathbb{R}^n$.

Let's keep $B = \{u_1, \ldots, u_n\}$

Let's rewrite $A = PDP^T$, by multiplying on the left by $P^{-1} = P^T$ and on the right by P

We obtain
$$P^T A P = \cancel{P^{-1}P} D \cancel{P^{-1}P}$$
$$= D.$$

Recall $Q(v) = v^T A v \quad \forall v \in \mathbb{R}^n$.

Let's make a "change of variable" by setting

$$y = P^{-1} v = P^T v, \text{ or equivalently}$$

$$v = Py.$$

Then the quadratic form may be written as
$$Q(v) = (Py)^T A (Py).$$
$$= y^T P^T A P y$$
$$= y^T D y \quad \text{(from above).}$$
$$= y^T \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix} y. \quad (**)$$

Let's interpret this change of variable.

Recall that $P = [u_1 \cdots u_n] = \underset{\mathcal{E} \leftarrow B}{P}$

and so $v = Py = \underset{\mathcal{E} \leftarrow B}{P}\, y$.

Moreover, $v = [v]_{\mathcal{E}} = \underset{\mathcal{E} \leftarrow B}{P}\, [v]_B$ from

So $$P\,[v]_B = v = Py$$

$$\Rightarrow \quad y = [v]_B.$$

Finally (**) takes the form

$$Q(v) = [v]_B^T \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix} [v]_B. \quad \forall\, v \in \mathbb{R}^n.$$

Theorem 4 Principal Axes Theorem

Let $A \in M_{n\times n}$ be symmetric and $A = PDP^T$ be an orthogonal diagonalization. Then

$$Q(v) = v^T A v = [v]_B^T\, D\, [v]_B, \quad \forall\, v \in \mathbb{R}^n$$

or equivalently

$$Q(v) = y^T D y$$

where $y = P^{-1} v$.